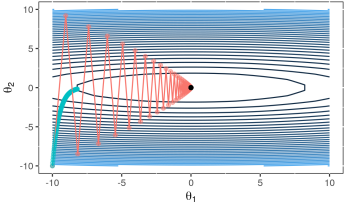
# Optimization in Machine Learning

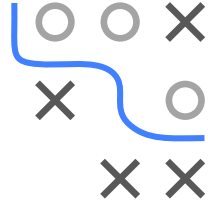
## First order methods
## Step size and optimality




**Learning goals**

- Impact of step size
- Fixed vs. adaptive step size
- Exact line search
- Armijo rule & Backtracking
- Bracketing & Pinpointing

# CONTROLLING STEP SIZE: FIXED & ADAPTIVE

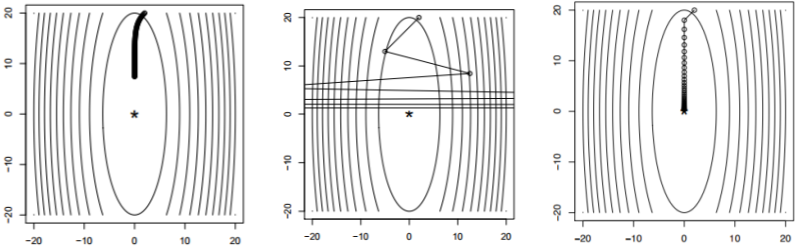


Iteration $t$: Choose not only descent direction $\mathbf{d}^{[t]}$, but also step size $\alpha^{[t]}$

First approach: **Fixed** step size $\alpha^{[t]} = \alpha > 0$

- If $\alpha$ too small, procedure may converge very slowly (left)
- If $\alpha$ too large, procedure may not converge $\rightarrow$ "jumps" around optimum (middle)

**Adaptive** step size $\alpha^{[t]}$ can provide better convergence (right)




Steps of line searches for $f(\boldsymbol{x}) = 10x_1^2 + x_2^2/2$

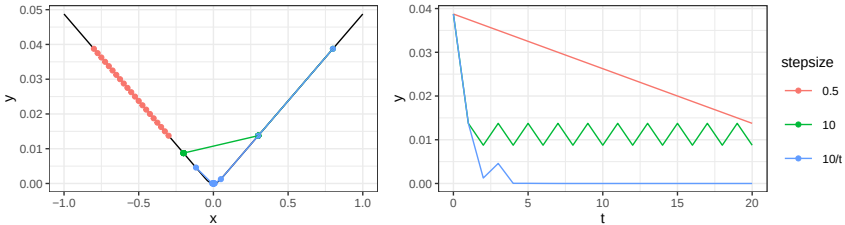
# STEP SIZE CONTROL: DIMINISHING STEP SIZE

How can we adaptively control step size?

A natural way of selecting $\alpha^{[t]}$ is to decrease its value over time

**Example:** GD on

$$f(x) = \begin{cases} \frac{1}{2}x^2 & \text{if } |x| \leq \delta, \\ \delta \cdot (|x| - 1/2 \cdot \delta) & \text{otherwise.} \end{cases}$$




GD with small constant (**red**), large constant (**green**), and diminishing (**blue**) step size

# STEP SIZE CONTROL: EXACT LINE SEARCH

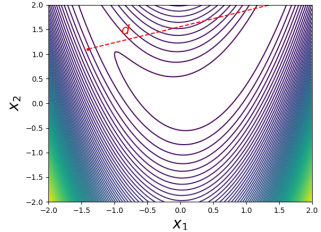
Use **optimal** step size in each iteration:

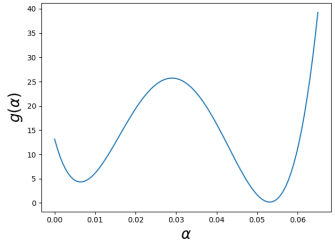
$$\alpha^{[t]} = \underset{\alpha \in \mathbb{R}_{\geq 0}}{\arg\min}\, g(\alpha) = \underset{\alpha \in \mathbb{R}_{\geq 0}}{\arg\min}\, f(\boldsymbol{x}^{[t]} + \alpha \mathbf{d}^{[t]})$$

Need to solve a **univariate** optimization problem in each iteration
$\Rightarrow$ univariate optimization methods

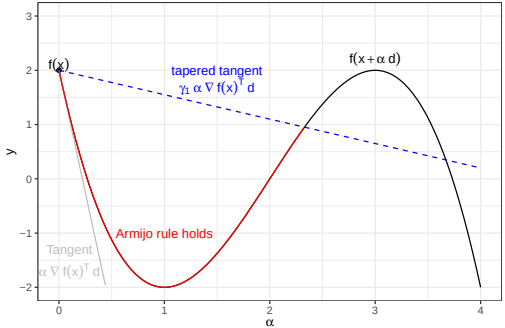
**Problem:** Expensive, prone to poorly conditioned problems

**But:** No need for *optimal* step size. Only need a step size that is "good enough".
**Reason:** Effort may not pay off, but in some cases slows down performance.



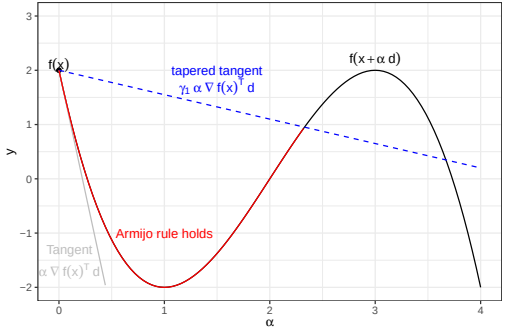
# ARMIJO RULE




**Inexact line search:** Minimize objective “sufficiently” without computing optimal step size exactly

Common condition to guarantee “sufficient” decrease: **Armijo rule**

# ARMIJO RULE

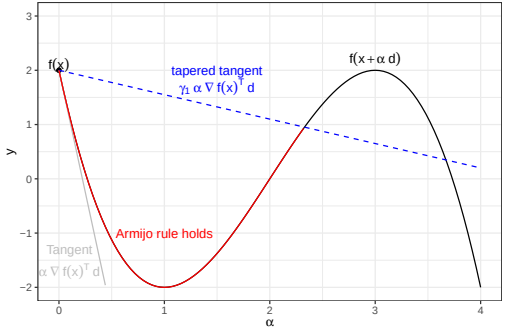



Fix $\gamma_1 \in (0, 1)$. $\alpha$ satisfies **Armijo rule** in $\mathbf{x}$ for descent direction $\mathbf{d}$ if

$$f(\mathbf{x} + \alpha\mathbf{d}) \leq f(\mathbf{x}) + \gamma_1 \alpha \nabla f(\mathbf{x})^\top \mathbf{d}.$$

**Note:** $\nabla f(\mathbf{x})^\top \mathbf{d} < 0$ ($\mathbf{d}$ *descent* dir.) $\implies f(\mathbf{x} + \alpha\mathbf{d}) < f(\mathbf{x})$.

# ARMIJO RULE




**Feasibility:** For descent direction **d** and $\gamma_1 \in (0, 1)$, there exists $\alpha > 0$ fulfilling Armijo rule. In many cases, Armijo rule guarantees local convergence of GD and is therefore frequently used.

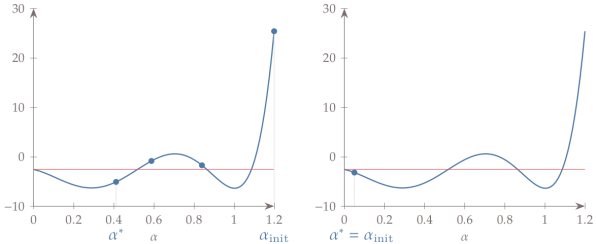
# BACKTRACKING LINE SEARCH

Procedure to meet the Armijo rule: **Backtracking** line search

**Idea:** Decrease $\alpha$ until Armijo rule is met

**Algorithm** Backtracking line search

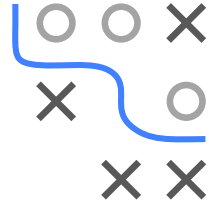
1: Choose initial step size $\alpha = \alpha_{\text{init}}$, $0 < \gamma_1 < 1$ and $0 < \tau < 1$
2: **while** $f(\boldsymbol{x} + \alpha \mathbf{d}) > f(\boldsymbol{x}) + \gamma_1 \alpha \nabla f(\boldsymbol{x})^\top \mathbf{d}$ **do**
3: Decrease $\alpha$: $\alpha \leftarrow \tau \cdot \alpha$
4: **end while**




(Source: Martins and Ning. *Engineering Design Optimization*, 2021.)

# BACKTRACKING LINE SEARCH / 2

# WOLFE CONDITIONS



Backtracking is simple and shows good performance in practice

**But:** Two undesirable scenarios

1. Initial step size $\alpha_{\text{init}}$ is too large $\Rightarrow$ need multiple evaluations of $f$
2. Step size is too small with highly negative slopes

**Solution** for small step sizes:

- Fix $\gamma_2$ with $0 < \gamma_1 < \gamma_2 < 1$.
- $\alpha$ satisfies **sufficient curvature condition** in $\mathbf{x}$ for $\mathbf{d}$ if

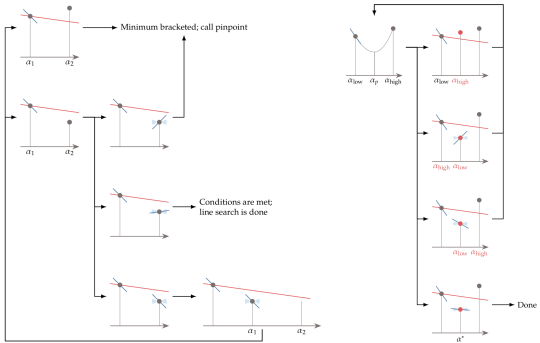
$$|\nabla f(\mathbf{x} + \alpha \mathbf{d})^\top \mathbf{d}| \leq \gamma_2 |\nabla f(\mathbf{x})^\top \mathbf{d}|.$$

Armijo rule + sufficient curvature condition = **Wolfe conditions**

# WOLFE CONDITIONS / 2

**Algorithm** for finding a Wolfe point (point satisfying Wolfe conditions):

1. **Bracketing:** Find interval containing Wolfe point
2. **Pinpointing:** Find Wolfe point in interval from bracketing



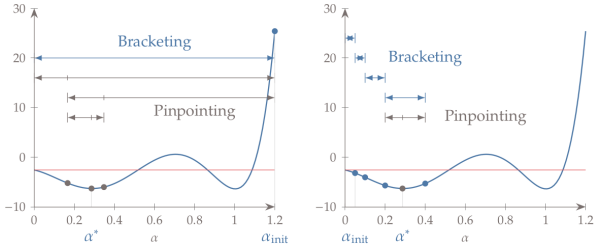

**Left:** Bracketing. **Right:** Pinpointing.

(Source: Martins and Ning. *EDO*, 2021.)

# BRACKETING & PINPOINTING

**Example:**

- Large initial step size results in quick bracketing but multiple pinpointing steps (**left**).
- Small initial step size results in multiple bracketing steps but quick pinpointing (**right**).




Source: Martins and Ning. *EDO*, 2021.